

02LQD Specification and Simulation of Digital Systems

2019/01/28

1. Combinational circuit design (3 points)

Binary coded decimal (BCD) encoding assigns a 4-bit binary code to each digit from 0 through 9 in base 10. The 4-bit BCD code for any particular single base-10 digit is its representation in binary notation. The 9-complement of a single BCD digit d is defined as $9-d$ by the following table :

BCD digit d	9-complement of d
0000	1001
0001	1000
0010	0111
0011	0110
0100	0101
0101	0100
0110	0011
0111	0010
1000	0001
1001	0000

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (7 points)

A Mealy machine acts as a recognizer of sequences transmitted on a serial line. It has 2 inputs and 2 outputs:

- the 1-bit input signal En enables machine operation when it is 1. Whenever it is 0 the machine returns to its initial state and waits for a new sequence
- the 1-bit input signal X is the serial line
- the 2-bit output signal $Z=Z_1Z_0$ is 10 when the input sequence 111 has been recognized on the last 3 bits, 01 when the input sequence 00 has been recognized on the last 2 bits, 00 otherwise.

Reset is synchronous.

Example

```
En  001111111111111110
X    010100011010011111
Z1 000000000000000110
Z0 000001100000100000
```

1. Draw the state diagram (3 points)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).

3. FSM-D design (20 points)

Pell numbers are recursively defined as follows:

$$P_n = 2P_{n-1} + P_{n-2} \quad n > 1$$

$$P_0 = 0$$

$$P_1 = 1$$

The first 8 Pell numbers are thus: 0 1 2 5 12 29 70 169.

Design a FSM-D to **iteratively** compute Pell numbers of order n , where n is on 3 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready`

signal is asserted for one clock cycle. Reset is asynchronous. The output is P. Define the appropriate size for P. The output P makes sense only when ready is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (7 points)
2. Design a FSM-D (13 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram
 - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Thursday January 31 1.59 pm**
- format of .zip archive name surname_ID (example Rossi_244123)
 - remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
 - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.